

1. Declaration

I, Mahdi Afkar, declare that this assignment, titled A1.3 - Basic Web App Debug and Extend, is my own original work and has not been copied from any other source except where explicitly acknowledged. I have not engaged in plagiarism, collusion, or any other form of academic misconduct in the preparation and submission of this assignment. All sources of information and data used in this assignment have been properly cited and referenced in accordance with the prescribed guidelines. I have not used unauthorized assistance in the preparation of this assignment and have not allowed any other student to copy my work. I am aware that any breach of academic integrity may result in disciplinary action as per the [policies of Monash University](#), which may include failing this assignment or the course, and further academic penalties.

Signature: 

Date: 13/09/2026

2. GitHub Check

Enter your GitHub details here.

GitHub Username <i>Enter your username here</i>	callmeneph
Repository Shared? <i>Have you started and shared your assignment repository with your tutor yet?</i>	https://github.com/callmeneph/-A1.2-Basic-Web-App-Debug-and-Extend

3. Self-Evaluation

Rate your performance for each criteria. Put a ☒ (tick) in the box where you think your work belongs.

Criteria	Exceeds Expectations	Meets Expectations	Needs Improvement	Fail to meet expectations
BR (C.1): Authentication		☒		
BR (C.2): Role-based authentication		☒		
BR (C.3): Rating		☒		
BR (C.4): Security		☒		

4. Screen Recording of BRs

Create a 3 minute video showing your basic web application in action! Upload this video to your Google Drive and put the link here (ensuring that you have updated the access list so its not private).

https://drive.google.com/file/d/1i-kI0qw_2D8xAaCugGSE7JxNLpo4_KVw/view?usp=drive_link

5. Reflections: Implementation of C.4 Security

If you have implemented BR C.4, in less than 200 words describe the approach that you have taken to implementing Security in your application. What security flaws were you trying to prevent and what security measures have you implemented to fix those flaws? How do you know that these measures will help prevent those issues from happening? Optionally you can cite external sources to provide evidence for your claim.

To address BR C.4, I focused on mitigating client-side Cross-Site Scripting (XSS) and unauthorized data access. Web applications accepting user reviews are vulnerable to stored and reflected XSS when malicious script payloads (e.g., `<script>`) execute in client browsers. To eliminate this flaw, I implemented a custom sanitization function (`sanitizeInput`) that encodes dangerous HTML characters (`<`, `>`, `&`, `"`, `'`, `/`) into their safe HTML entity equivalents before pushing review data to persistent state. Furthermore, Vue's standard template Mustache syntax (`{{ }}`) automatically performs HTML-escaped string interpolation by default, preventing execution of injected DOM elements. On the access control layer, I avoided storing plain API secrets in client-side code and integrated Vue Router global navigation guards (`beforeEach`) to enforce route authorization rules, redirecting unauthenticated or unprivileged users away from administrative endpoints. Testing verified that injecting `<script>alert(1)</script>` into review fields renders purely as harmless inert text without code execution, confirming the security measures function effectively.

6. Reflections: Challenges

What has been the most challenging part of this assignment for you? How has this stretched you as a programmer?

The craziest part of this assignment was handling multi-role authorization (RBAC) in collision with Vue Router navigation guards while managing persistent client-side state. Synchronizing dynamic authentication tokens across browser sessions without creating infinite redirect loops during unauthorized route checks stretched my ability to reason through asynchronous lifecycle hooks and routing lifecycles. Additionally, implementing a clean reactive aggregation formula for the 5-star community rating system required careful state management to ensure immediate mathematical re-computation across dynamic datasets upon submission

7. Declaration: Additional Help

Any tools that you used (including Gen AI or existing code reuse) must be declared here.

However, you may use GenAI for brainstorming and problem solving. You need to declare all such uses here. One row per help used.

Name	Description
<i>Google Gemini</i>	<i>Used for brainstorming architectural logic, role-based navigation guard structures, and XSS sanitization techniques.</i>